

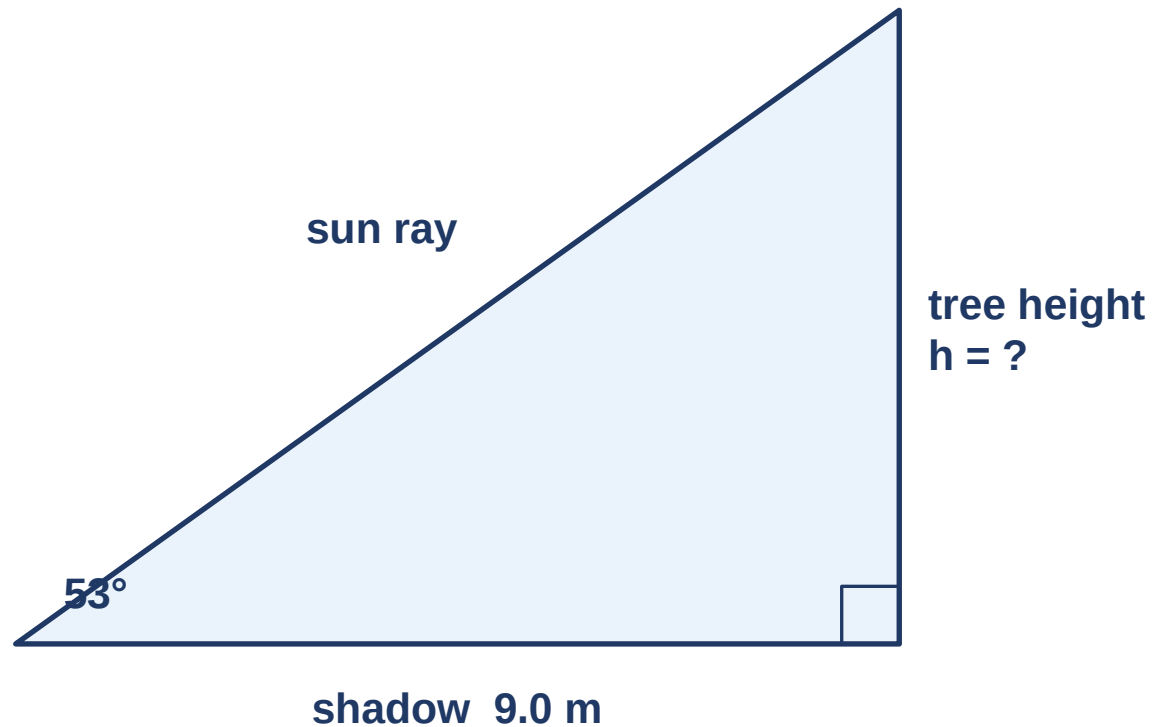
MATHEMATICS — GRADE 10 | TRIGONOMETRY I

The Tangent Ratio

Solving the mango-tree problem

Today we calculate the height of the Kisumu mango tree for the first time.

How tall is the mango tree?



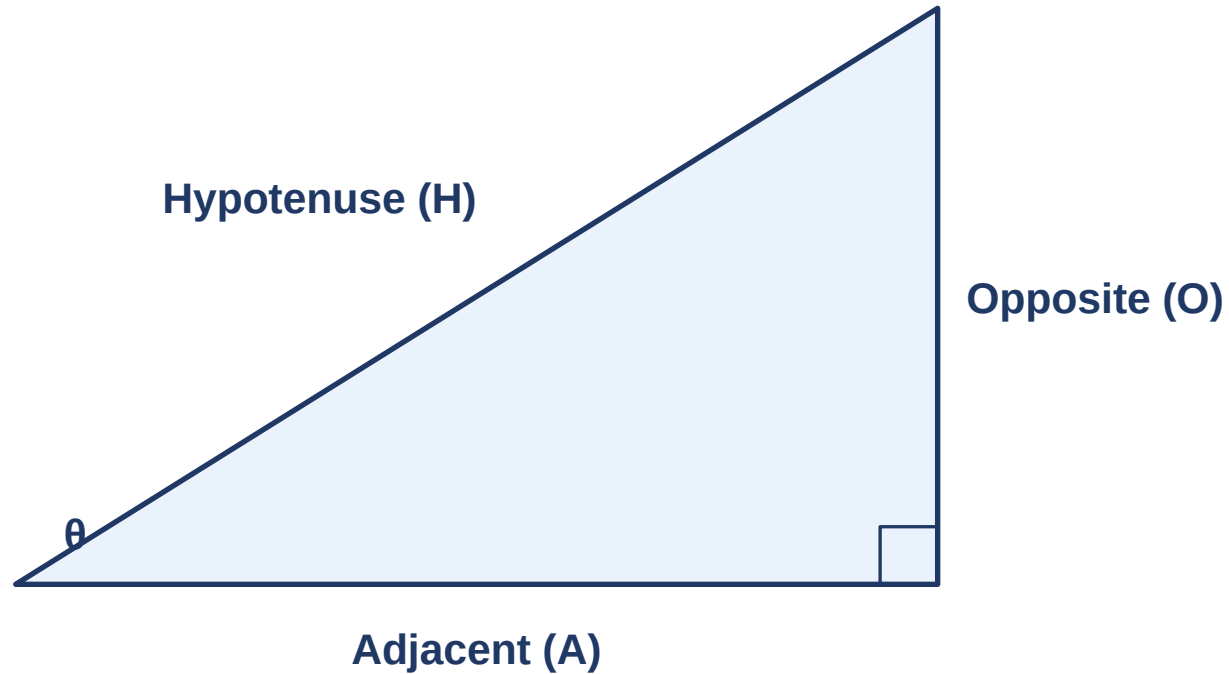
A student in Kisumu measures:

- Sun angle: 53°
- Tree's shadow: 9.0 m
- From last lesson: $\sin 53^\circ \approx 0.799$, $\cos 53^\circ \approx 0.602$

Predict: which ratio will help most? Estimate the height.

Write your prediction silently (2 minutes), then share with your partner.

Sine and cosine compare sides with the hypotenuse.



$$\sin \theta = O / H$$

$$\cos \theta = A / H$$

Opposite: the side facing the angle. Adjacent: the side next to the angle (not the hypotenuse).

Problem: the hypotenuse is the sun ray. We can't measure it with a tape measure. We need a ratio that skips it.

Divide sine by cosine and watch H disappear.

1. $\sin \theta \div \cos \theta = (O/H) \div (A/H)$
2. $= (O/H) \times (H/A)$ (dividing by a fraction = multiplying by its reciprocal)
3. $= O / A$ (the H's cancel)
4. So $\tan \theta = \sin \theta / \cos \theta = \text{Opposite} / \text{Adjacent}$

Copy each step and draw arrows showing where H cancels. Memory aid: TOA — Tan = Opposite / Adjacent.

QUICK CHECK 1

In a right-angled triangle, which ratio equals $\tan \theta$?

A Opposite / Hypotenuse

B Adjacent / Hypotenuse

C Opposite / Adjacent

D Hypotenuse / Opposite

Answer key is teacher-side only, in a separate file.

Does $\sin \theta \div \cos \theta$ really equal $\tan \theta$?

θ	$\sin \theta$	$\cos \theta$	$\sin \theta \div \cos \theta$	$\tan \theta$ (tables)	Match?
30°	0.5000	0.8660	0.5774	0.5774	✓
40°	0.6428	0.7660	0.8391	0.8391	✓
60°					
70°					

Your turn: complete the rows for 60° and 70° using your mathematical tables. Small differences in the last digit come from rounding.

QUICK CHECK 2

$\sin 60^\circ = 0.8660$ and $\cos 60^\circ = 0.5000$. Use $\tan \theta = \sin \theta / \cos \theta$ to find $\tan 60^\circ$.

A 0.5774

B 1.7320

C 0.4330

D 1.3660

Answer key is teacher-side only, in a separate file.

Back to the mango tree

Which side is 9.0 m?

Stand at the shadow tip and look at the 53° angle. Is the shadow the side facing the angle, or the side next to it?

What are we looking for?

The tree height h faces the 53° angle. Which ratio uses both the side we know and the side we want — without the hypotenuse?

30 seconds of silent thinking, then discuss in groups of four.

QUICK CHECK 3

Relative to the 53° angle, the 9.0 m shadow is the triangle's ...

A Opposite side

B Adjacent side

C Hypotenuse

D It depends on how tall the tree is

Answer key is teacher-side only, in a separate file.

The mango tree is about 12 metres tall.

Method 1: tangent ratio

$$\tan 53^\circ = h / 9.0$$

$$h = 9.0 \times \tan 53^\circ$$

$$h = 9.0 \times 1.3270$$

$$h \approx \mathbf{11.94 \text{ m}}$$

Method 2: similar triangles (earlier lessons)

Student: 1.6 m tall, 1.2 m shadow

$$\text{Ratio height} \div \text{shadow} = 1.6 \div 1.2 \approx 1.333$$

$$h = 9.0 \times 1.333$$

$$h \approx \mathbf{12.0 \text{ m}}$$

Both methods agree (to rounding). $\tan \theta$ is the height-to-shadow ratio for that sun angle — it is the same for every object at the same time of day.

QUICK CHECK 4

$\tan 53^\circ = h / 9.0$. Which step correctly finds h ?

A $h = 9.0 \div \tan 53^\circ$

B $h = 9.0 \times \tan 53^\circ$

C $h = \tan 53^\circ \div 9.0$

D $h = 9.0 + \tan 53^\circ$

Answer key is teacher-side only, in a separate file.

Why is tan so useful for measuring heights in Kenya?

- tan uses the opposite and adjacent sides — a height you want and a ground distance you can pace or tape.
- No hypotenuse needed: you never have to measure along a sun ray or up a slope.
- Surveyors, builders and farmers use this with one angle and one ground measurement.

Try it: a flagpole casts a 10 m shadow when the sun angle is 40° ($\tan 40^\circ = 0.8391$). How tall is it?

QUICK CHECK 5

A flagpole casts a 10 m shadow when the sun angle is 40° . $\tan 40^\circ = 0.8391$. How tall is the flagpole?

A 0.84 m

B 8.39 m

C 10.84 m

D 11.92 m

Answer key is teacher-side only, in a separate file.

Update your model. Add one new question.

UPDATE YOUR MODEL

On your trigonometry model triangle: label $\theta = 53^\circ$, adjacent = 9.0 m (shadow), opposite = h (tree). Write $\tan 53^\circ = h / 9.0$ inside it. Add a “ratio web” linking $\sin \theta$, $\cos \theta$ and $\tan \theta$ with the identity $\tan \theta = \sin \theta / \cos \theta$.

DRIVING QUESTION BOARD

Three columns on the board:

ANSWERED TODAY: “We can now find the tree’s height because ...”

STILL WONDERING: e.g. what if we know the height but not the angle?

CONNECTIONS: how does tan link to Lessons 1–3?

From the ARES offline library



VIDEO [Trigonometry Word Problems Principles - Basic](#)

CK-12 • Good match: right-triangle word problems



READING [Trigonometry Word Problems \(Flexbook\)](#)

CK-12 • Good match: worked examples using $\tan$, $\sin$ and $\cos$



SEARCH [Search ARES: trigonometry, tangent ratio](#)

Opens an ARES search for related videos and notes

Links open only on the school network (ares.local). Click to open during the lesson.

5 questions asked during class today

1. Which ratio equals $\tan \theta$?
2. $\sin 60^\circ = 0.8660$, $\cos 60^\circ = 0.5000$. Find $\tan 60^\circ$.
3. Relative to the 53° angle, the 9.0 m shadow is the ...
4. $\tan 53^\circ = h / 9.0$. Which step finds h ?
5. Flagpole: 10 m shadow, sun angle 40° ($\tan 40^\circ = 0.8391$). Height?

Ask your teacher how to submit your answers today (for example, on a paper answer slip).